Machine Learning Assignment :- 6

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Use the diabetes data set from UCI and Pima Indians Diabetes data set for performing the following:

a. Univariate analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis

b. Bivariate analysis: Linear and logistic regression modeling

c. Multiple Regression analysis

d. Also compare the results of the above analysis for the two data sets. Dataset link: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

Imported Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
from sklearn.datasets import load_diabetes
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, confusion_matrix
```

```
df = pd.read_csv('diabetes.csv')

df = df.iloc[:, 1:]

cols_to_fix = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']
df[cols_to_fix] = df[cols_to_fix].replace(0, np.nan)
df = df.dropna()

df.to_csv('cleaned_diabetes.csv', index=False)
```

```
dp = pd.read_csv('cleaned_diabetes.csv')
dp
```

	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	89.0	66.0	23.0	94.0	28.1	0.167	21	0
1	137.0	40.0	35.0	168.0	43.1	2.288	33	1
2	78.0	50.0	32.0	88.0	31.0	0.248	26	1
3	197.0	70.0	45.0	543.0	30.5	0.158	53	1
4	189.0	60.0	23.0	846.0	30.1	0.398	59	1
...	...	...	...	...	...	...	...	...
387	181.0	88.0	44.0	510.0	43.3	0.222	26	1
388	128.0	88.0	39.0	110.0	36.5	1.057	37	1
389	88.0	58.0	26.0	16.0	28.4	0.766	22	0
390	101.0	76.0	48.0	180.0	32.9	0.171	63	0
391	121.0	72.0	23.0	112.0	26.2	0.245	30	0

392 rows × 8 columns

Next steps: Generate code with dp New interactive sheet

```
dp.shape
```

```
(392, 8)
```

```
dp.describe()
```

	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	392.000000	392.000000	392.000000	392.000000	392.000000	392.000000	392.000000	392.000000
mean	122.627551	70.663265	29.145408	156.056122	33.086224	0.523046	30.864796	0.331633
std	30.860781	12.496092	10.516424	118.841690	7.027659	0.345488	10.200777	0.471401
min	56.000000	24.000000	7.000000	14.000000	18.200000	0.085000	21.000000	0.000000
25%	99.000000	62.000000	21.000000	76.750000	28.400000	0.269750	23.000000	0.000000
50%	119.000000	70.000000	29.000000	125.500000	33.200000	0.449500	27.000000	0.000000
75%	143.000000	78.000000	37.000000	190.000000	37.100000	0.687000	36.000000	1.000000
max	198.000000	110.000000	63.000000	846.000000	67.100000	2.420000	81.000000	1.000000

corelation check

```
corelation_check = dp.corr()

print(corelation_check)


# print("\nCorrelation with Outcome:")
# print(df.corr()['Outcome'].sort_values(ascending=False))
```

```
                          Glucose  BloodPressure  SkinThickness   Insulin  \
Glucose                  1.000000       0.210027       0.198856  0.581223
BloodPressure            0.210027       1.000000       0.232571  0.098512
SkinThickness            0.198856       0.232571       1.000000  0.182199
Insulin                  0.581223       0.098512       0.182199  1.000000
BMI                      0.209516       0.304403       0.664355  0.226397
DiabetesPedigreeFunction 0.140180      -0.015971       0.160499  0.135906
Age                      0.343641       0.300039       0.167761  0.217082
Outcome                  0.515703       0.192673       0.255936  0.301429

                              BMI  DiabetesPedigreeFunction       Age  \
Glucose                  0.209516                  0.140180  0.343641
BloodPressure            0.304403                 -0.015971  0.300039
SkinThickness            0.664355                  0.160499  0.167761
Insulin                  0.226397                  0.135906  0.217082
BMI                      1.000000                  0.158771  0.069814
DiabetesPedigreeFunction 0.158771                  1.000000  0.085029
Age                      0.069814                  0.085029  1.000000
Outcome                  0.270118                  0.209330  0.350804

                          Outcome
Glucose                  0.515703
BloodPressure            0.192673
SkinThickness            0.255936
Insulin                  0.301429
BMI                      0.270118
DiabetesPedigreeFunction 0.209330
Age                      0.350804
Outcome                  1.000000
```

```
#Frequency of outcome column
dp['Outcome'].value_counts()
```

Outcome	count
0	262
1	130

dtype: int64

calculate statistical summary

```
from scipy.stats import skew,kurtosis

uni = pd.DataFrame({
    "Mean": dp.mean()
```

```
    "Median": dp.median(),
    "Mode": dp.mode().iloc[0],
    "Variance": dp.var(),
    "Std_Dev": dp.std(),
    "Skewness": dp.apply(skew),
    "Kurtosis": dp.apply(kurtosis)
})

print("Univariate Statistics Summary")
display(uni)
```

```
Univariate Statistics Summary
```

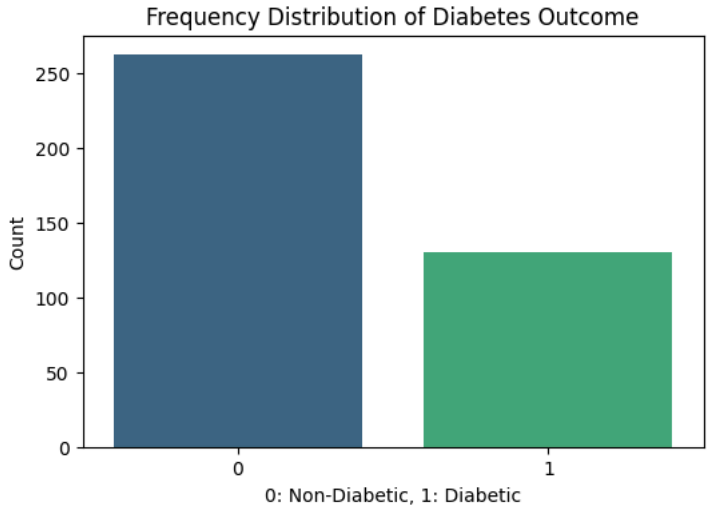
	Mean	Median	Mode	Variance	Std_Dev	Skewness	Kurtosis
Glucose	122.627551	119.0000	100.000	952.387781	30.860781	0.515866	-0.492353
BloodPressure	70.663265	70.0000	70.000	156.152304	12.496092	-0.087181	0.770028
SkinThickness	29.145408	29.0000	32.000	110.595171	10.516424	0.208509	-0.467146
Insulin	156.056122	125.5000	105.000	14123.347226	118.841690	2.156822	6.260449
BMI	33.086224	33.2000	32.000	49.387994	7.027659	0.660944	1.521463
DiabetesPedigreeFunction	0.523046	0.4495	0.692	0.119362	0.345488	1.951597	6.270504
Age	30.864796	27.0000	22.000	104.055842	10.200777	1.398230	1.700180
Outcome	0.331633	0.0000	0.000	0.222219	0.471401	0.715240	-1.488432

Next steps: Generate code with uni New interactive sheet

Frequency comparison

```
# # Frequency Analysis of Outcome
# print("Frequency of Diabetic vs Non-Diabetic Cases:")
# print(dp['Outcome'].value_counts())

# plt.figure(figsize=(6, 4))
# sns.countplot(x='Outcome', data=dp, palette='viridis')
# plt.title('Frequency Distribution of Diabetes Outcome')
# plt.xlabel('0: Non-Diabetic, 1: Diabetic')
# plt.ylabel('Count')
# plt.show()



import seaborn as sns  # This defines 'sns' to fix your error
import matplotlib.pyplot as plt

# Frequency Analysis of Outcome
print("Frequency of Diabetic vs Non-Diabetic Cases:")
print(dp['Outcome'].value_counts())

plt.figure(figsize=(6, 4))
# Now sns.countplot will work perfectly
sns.countplot(x='Outcome', data=dp, palette='viridis')
plt.title('Frequency Distribution of Diabetes Outcome')
plt.xlabel('0: Non-Diabetic, 1: Diabetic')
plt.ylabel('Count')
plt.show()
```

```
Frequency of Diabetic vs Non-Diabetic Cases:
Outcome
0    262
1    130
Name: count, dtype: int64
/tmp/ipykernel_2702/759889634.py:23: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

  sns.countplot(x='Outcome', data=dp, palette='viridis')
```

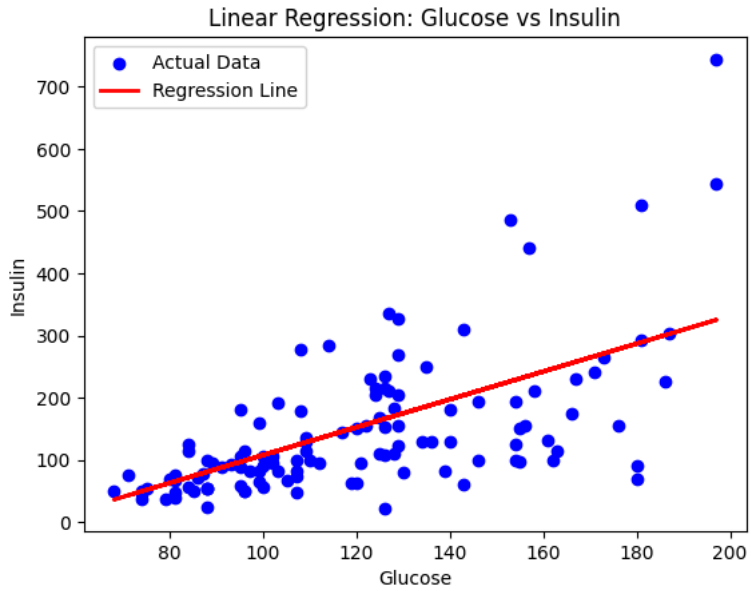


Single feature Linear regression

```
X = dp[['Glucose']]   # Single feature
y = dp[['Insulin']]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

lin_reg = LinearRegression()
lin_reg.fit(X_train, y_train)

y_pred = lin_reg.predict(X_test)

print("Linear Regression Results (Sklearn Dataset)")
print("Coefficient:", lin_reg.coef_)
print("Intercept:", lin_reg.intercept_)
print("R2 Score:", r2_score(y_test, y_pred))
print("MSE:", mean_squared_error(y_test, y_pred))
```

```
Linear Regression Results (Sklearn Dataset)
Coefficient: [[2.23986883]]
Intercept: [-116.25794942]
R2 Score: 0.38123642952556447
MSE: 7877.785642299832
```

```
import matplotlib.pyplot as plt
import seaborn as sns


plt.scatter(X_test, y_test, color='blue', label='Actual Data')


plt.plot(X_test, y_pred, color='red', linewidth=2, label='Regression Line')

plt.xlabel('Glucose')
plt.ylabel('Insulin')
plt.title('Linear Regression: Glucose vs Insulin')
plt.legend()
plt.show()
```



Logistic Regression

```
## --- SECTION: Bivariate Analysis (Logistic Regression) ---
from sklearn.linear_model import LogisticRegression

# Using Glucose to predict the Outcome (Diabetes or not)
X_log = dp[['Glucose']]
y_log = dp['Outcome']

X_train_l, X_test_l, y_train_l, y_test_l = train_test_split(X_log, y_log, test_size=0.3, random_state=42)

log_model = LogisticRegression()
log_model.fit(X_train_l, y_train_l)
y_pred_log = log_model.predict(X_test_l)

print("Logistic Regression (Glucose vs Outcome):")
print("Accuracy Score:", accuracy_score(y_test_l, y_pred_log))
print("Confusion Matrix:\n", confusion_matrix(y_test_l, y_pred_log))
```

```
Logistic Regression (Glucose vs Outcome):
Accuracy Score: 0.7203389830508474
Confusion Matrix:
 [[68 12]
 [21 17]]
```

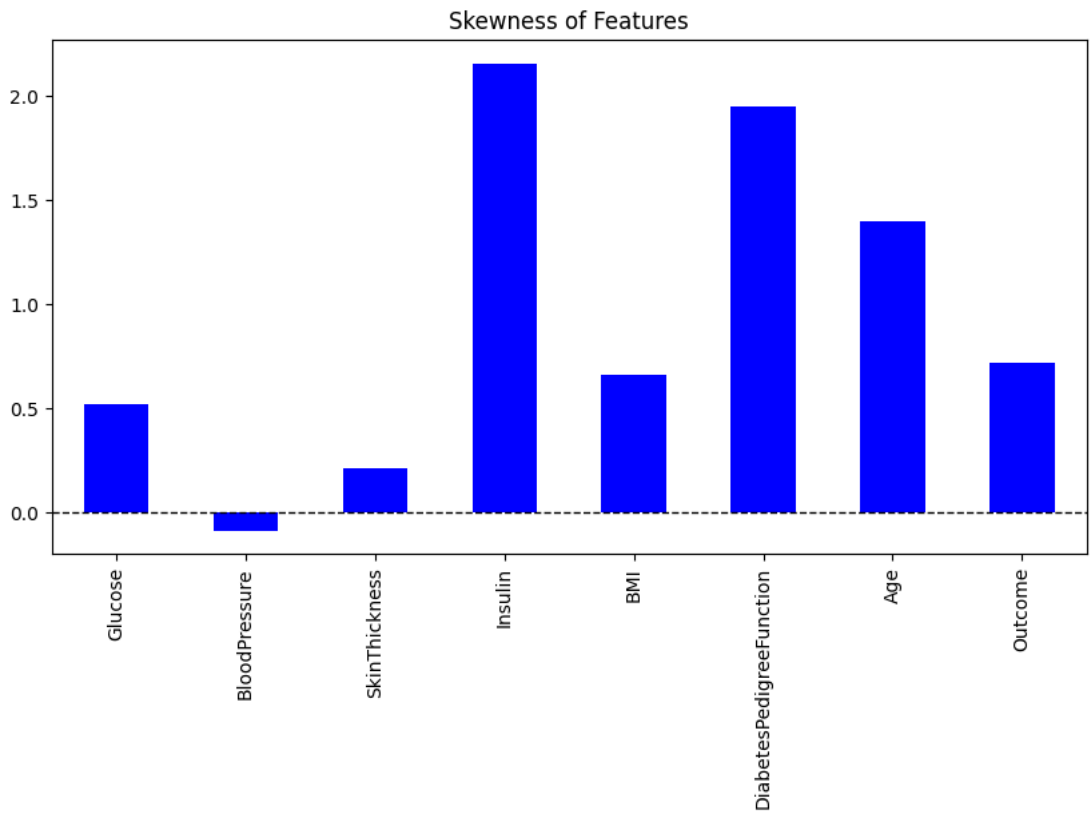
Multi feature Linear Regression

```
## --- SECTION: Multiple Regression Analysis ---
# Predicting Insulin using three features: Glucose, BMI, and Age
X_multi = dp[['Glucose', 'BMI', 'Age']]
y_multi = dp['Insulin']

X_train_m, X_test_m, y_train_m, y_test_m = train_test_split(X_multi, y_multi, test_size=0.3, random_state=42)

multi_model = LinearRegression()
multi_model.fit(X_train_m, y_train_m)
y_pred_m = multi_model.predict(X_test_m)

print("Multiple Regression Results (Predicting Insulin):")
print("R2 Score:", r2_score(y_test_m, y_pred_m))
print("MSE:", mean_squared_error(y_test_m, y_pred_m))
```

```
Multiple Regression Results (Predicting Insulin):
R2 Score: 0.38327114797881334
MSE: 7851.880633372372
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# skewness in uni
plt.figure(figsize=(10, 5))
uni['Skewness'].plot(kind='bar', color='blue')
```

```
plt.title('Skewness of Features')
plt.axhline(0, color='black', linestyle='--', linewidth=1)
plt.show()
```



Skewness curves

```
import seaborn as sns
import matplotlib.pyplot as plt

for col in dp.columns:
    sns.histplot(df[col], kde=True, color='skyblue')
    plt.title(f'Skewness of {col}')
    plt.show()
```